Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *French Pharmaceutical Regulatory Affairs — Document Compliance NLP*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark and deployment are both text-only in standard written French using the Latin alphabet with French diacritics; there is no script mismatch, no cross-modality gap, and no infrastructure mismatch. The benchmark uses HuggingFace Datasets/Transformers/PyTorch [Q48] consistent with the deployment's likely model integration. Tokenization analysis is documented [Q52, Q53, Q78–Q80], providing tokenizer-agnostic evaluation infrastructure relevant to handling long INN strings and pharmaceutical compound terms. Two minor caveats: PxCorpus originates from spoken-language transcription with disfluencies, code-switching, and artifacts (PXCORPUS-D12, PXCORPUS-D13, PXCORPUS-D14, PXCORPUS-D15) that diverge from regulatory document register, and E3C contains 8 of 10 configurations in non-French languages (E3C-D6, E3C-D7, E3C-D8) requiring careful configuration filtering. These do not undermine signal-distribution alignment for the metropolitan French deployment but warrant care when constructing fine-tuning subsets.

ID	Evidence
Q78	“However, our experiments, as shown in Table 7, reveal that FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average).” (p.8)
Q79	“Surprisingly, when comparing the performance of these two models on the benchmark tasks, we observe that DrBERT-CP outperforms FlauBERT on 16 out of the 20 tasks, thus contradicting previous conclusions drawn by the community.” (p.8)
Q80	“Yet, tokenization, as it is currently done in MLMs, seems to play a minor role in the performance of systems.” (p.8)
Q48	“We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries.” (p.5)
Q52	“To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023).” (p.6)
Q53	“It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process.” (p.6)
PXCORPUS-D12	/chet — Single-token transcription artifact; no pharmaceutical content
PXCORPUS-D13	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français... — Code-switched colloquial fragment from dialogue session
PXCORPUS-D14	ne pas tenir compte à midi tous les jours merd/ — Spoken correction with truncation artifact
PXCORPUS-D15	euh 2 le matin 2 le midi et 2 le soir — Spoken filler with no drug name; no regulatory document parallel
E3C-D6	Azken hilabeteetan Ikernek kodeina + ibuprofenoa hartu ditu, baina ez du hobekuntza handirik nabaritu — Basque text; irrelevant to French deployment
E3C-D7	A chest radiograph showed right upper lobe consolidation with volume loss, right para-tracheal and left hilar adenopathy — English clinical text; irrelevant to French regulatory deployment
E3C-D8	Trattata con nutrizione parenterale(NP), volume iniziale di 240 ml/die(166 kcal) — Italian clinical text; irrelevant to French regulatory deployment

Strengths

- Both benchmark and deployment are text-only in standard written French with Latin script — no signal-level mismatch [Q31, Q48]
- HuggingFace toolkit ecosystem matches deployment infrastructure assumptions [Q48]
- Tokenizer-agnostic evaluation via SeqEval IOB2 first-subtoken protocol [Q52, Q53] supports robust evaluation of long pharmaceutical compound terms
- Documented tokenization sub-token statistics (1.43–1.90 sub-tokens/word) [Q78] provide a baseline for assessing pharmaceutical-term fragmentation

ID	Evidence
Q31	“While the dataset covers 5 languages, only the French portion is retained for the benchmark.” (p.3)
Q48	“We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries.” (p.5)
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match: standard written French text in both benchmark and deployment; no resolution, modality, or capture-parameter divergence applicable to text [Q31, Q48].

IF-2: Yes — metropolitan French digital infrastructure supports the same text data formats; high-resource context consistent across benchmark and deployment.

IF-3: Domain-specific form differences are minor: PxCorpus spoken transcription register differs from formal regulatory document register (PXCORPUS-D12, PXCORPUS-D14); MANTRAGSC fr_patents uses patent claim register distinct from PIL register (MANTRAGSC-D14); fr_medline uses title-only fragments. These can be filtered when assembling fine-tuning subsets.

IF-4: No major form mismatches threaten external validity for the text-only metropolitan-French deployment. Minor: E3C non-French configurations (E3C-D6, E3C-D7, E3C-D8) and MANTRAGSC German contamination (MANTRAGSC-D13) require configuration-level filtering.

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PXCORPUS-D12	/chet — Single-token transcription artifact; no pharmaceutical content

MANTRAGSC-D14	`"Composé selon l'une quelconque des revendications 3 à 8, dans lequel R106 est choisi parmi un hydrogène, un alkyle substitué ou insubstitué en C1-C8 linéaire ou ramifié"` — Chemical patent claim language; distinct from patient leaflet register
PXCORPUS-D14	ne pas tenir compte à midi tous les jours merd/ — Spoken correction with truncation artifact
E3C-D6	Azken hilabeteetan Ikernek kodeina + ibuprofenoa hartu ditu, baina ez du hobekuntza handirik nabaritu — Basque text; irrelevant to French deployment
MANTRAGSC-D13	`"Die empfohlene Dosis für Agenerase Lösung beträgt 17 mg... Amprenavir/kg Körpergewicht"` — German-language example; not relevant to French deployment
E3C-D7	A chest radiograph showed right upper lobe consolidation with volume loss, right para-tracheal and left hilar adenopathy — English clinical text; irrelevant to French regulatory deployment
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Information Gaps

- Sub-token fragmentation rates for INN strings and ATC codes specifically — benchmark reports only general averages